

Question 1:

1.
 - a) The hidden technical debt most related to this issue is Boundary erosion. In this case the Statistical logic of the “favorite restaurants” recommendation feature mixed with the rule based rounding off logic for “estimated delivery time” and the interface between the two broke down.
 - b) The hidden technical debt most related to this issue is undeclared consumers. In this case the marketing team is silently consuming the model outputs which creates invisible dependencies that the ML team is unaware of.
 - c) The hidden technical debt most related to this issue is Configuration and Glue code debt. The information across different training pipelines is unorganised and scattered across the 14 undocumented shell scripts creating tangled pipeline jungles.
2. We can resolve the technical debt in c) by orchestrating experiment tracking using Mlflow. It is a tracking server, store and registry which can organise the untracked shell scripts.